

# INSTANT + SCHEDULED ORDER ASSIGNMENT ENGINE

## Problem Statement

### What We're Solving

- Tenzo operates a home services marketplace where the same expert pool must serve two fundamentally different customer promises simultaneously, with no system to manage the conflict between them.
- Tenzo offers two order types: **Instant** (on-demand, help as soon as possible) and **Scheduled** (prepaid, future time slot). Experts work 10-hour shifts (8 AM–6 PM) with mandatory pre-travel (10 min), post-travel (10 min), and buffer time between jobs. Each expert can serve only one order at a time.
- The core problem is that scheduled orders are assigned only minutes before they start — leaving no time for failure recovery. Meanwhile, instant orders are accepted without checking whether they will conflict with upcoming scheduled slots. The result is a system where the two order types cannibalize the same expert pool without any prioritization logic, protection windows, or workload balancing.

### Current State — What's Broken

- × Scheduled orders assigned minutes before start — no recovery window
- × Instant orders accepted with no conflict check against upcoming slots
- × No workload balancing — some experts overloaded, others idle
- × Cancellations communicated reactively, never proactively
- × No priority hierarchy — all orders treated equally
- × No escalation path when assignment fails

### Business Impact

- 18% cancellation rate erodes customer lifetime value
- 84% OTF means 1 in 6 orders arrives late
- 52% instant acceptance rate — half of demand lost
- 68% expert utilization — significant revenue left on table
- Scheduled customers bear all risk of system failures
- No ops visibility until after the cancellation happens

### The Core Scenario

A scheduled order is set for 4:00 PM. An instant order arrives at 3:29 PM. Should the system accept the instant order? If it does, the expert may be unavailable for the 4:00 PM prepaid booking. If it rejects blindly, real-time demand is lost unnecessarily. The system currently has no logic to answer this — leading to either cascading cancellations or broad rejection of instant orders. Neither is acceptable at scale.

# Scope Definition

## ✓ Goals

- Prepaid orders confirmed no later than T-20 min. Travel and buffer windows are hard constraints, no two orders can ever share an expert.
- Show deferred ETAs instead of hard rejections. Score experts on workload, idle time, and shift capacity, not just proximity, so every available slot gets filled intelligently.
- Proactive communication on every delay and change. Real-time ops dashboard with automatic escalation flows so cancellations are prevented, not just responded to.

## ✗ Non-Goals

- Dynamic surge pricing and skill-based routing (deep clean vs. standard) are out of scope. This system assigns experts — it does not reprice orders or match by category.
- Team dispatch, multi-expert orders, and "return to my favourite expert" requests are not handled. Assignment is purely algorithmic based on availability and scoring.
- Live traffic-aware routing and variable shift scheduling are future enhancements. Travel times use a static distance matrix and shifts are fixed at 8 AM–6 PM.

| Metric                        | Current     | 90-Day Target | Measured By                                          |
|-------------------------------|-------------|---------------|------------------------------------------------------|
| Cancellation rate             | 18%         | <5%           | Orders cancelled / orders placed                     |
| On-Time Fulfillment (OTF)     | 84%         | >95%          | Expert arrival within ±5 min of promised time        |
| Instant acceptance rate       | 52%         | >80%          | (Accepted + deferred confirmed) / instant requests   |
| Scheduled fulfillment rate    | 88%         | >97%          | Scheduled orders completed on time / total scheduled |
| Expert utilization            | 68%         | >85%          | Active service time / total shift time               |
| Protection window breach rate | Not tracked | <2%           | Orders unassigned at T-20 / total scheduled          |

## Assumptions

Constraints we're designing within. If any of these change, the system design needs to be revisited.

| Assumption                                      | Value/Rule                                                                                   | Impact If Wrong                                                                          |
|-------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| All experts are generalists                     | Any expert can serve any order type or location                                              | Routing logic needs skill-matching layer; scoring formula changes significantly          |
| Travel matrix is static                         | Manikonda↔Narsingi=10 min,<br>Narsingi↔Puppalaguda=8 min,<br>Manikonda↔Puppalaguda=12 min    | Variable shift logic needed; capacity calculations change per expert                     |
| Shifts are fixed                                | 8:00 AM – 6:00 PM daily, all 4 experts                                                       | Team dispatch logic and concurrent assignment rules required                             |
| Scheduled orders are prepaid                    | Payment captured at booking time                                                             | If not prepaid, cancellation penalties change; priority hierarchy may need re-evaluation |
| Current time is 8:00 AM for scheduling exercise | All assignments calculated from this base time                                               | Real-time system would recalculate continuously throughout the shift                     |
| Buffer rules are consistent                     | Instant: 15 min post buffer.<br>Scheduled: 15 min post buffer.<br>Pre-travel: 10 min always. | Any rule change propagates through all protection window calculations                    |
| Experts start at their home location            | E1=Manikonda, E2=Narsingi, E3=Puppalaguda, E4=Manikonda                                      | If experts start elsewhere, initial travel calculations are inaccurate                   |

# COMPETITIVE BENCHMARKING

## How Others Solve This

How leading home services and on-demand platforms handle the instant vs. scheduled conflict — and what Tenzo can learn from each.

### Urban Company

#### Scheduling Approach

Scheduled-first model. Pre-assigns professionals 2-4 hours before service start. Instant available in select cities only.

#### Protection Window

Hard 2-hour pre-assignment lock. Professional cannot accept other work once assigned.

#### Instant Handling

Instant treated as a separate pool of "on-demand" professionals — not shared with scheduled.

#### What Tenzo Can Learn

✓ Separate pools prevent conflict.  
✗ Reduces utilization significantly — idle on-demand pros during low instant demand.

### Uber (rides)

#### Scheduling Approach

Purely instant — no scheduled conflict exists. But Uber Reserve (scheduled) uses a 15-min pre-dispatch window.

#### Protection Window

Driver dispatched 15 min before pickup. Not blocked earlier — accepted only when trip is imminent.

#### Instant Handling

Instant always wins; Reserve is dispatched only when driver is available and close enough.

#### What Tenzo Can Learn

✓ Short protection windows keep utilization high.  
✓ "Show later ETA" equivalent via surge pricing signals demand to drivers.

### Swiggy / Zomato

#### Scheduling Approach

Purely instant food delivery — no scheduled conflict. Scheduled delivery (pre-orders) is a recent addition.

#### Protection Window

Pre-order assigned 30-40 min before slot via dynamic batching algorithm.

#### Instant Handling

Instant orders dynamically batched. Pre-orders frozen from instant batching 30 min before.

#### What Tenzo Can Learn

✓ Batching concept applicable — Tenzo could "freeze" expert from instant pool 30 min before scheduled start.

### Dunzo / Zepto

#### Scheduling Approach

Instant only. No scheduled model currently.

#### Protection Window

N/A

#### Instant Handling

Zone-based assignment. Delivery partners assigned to zones, not individual orders, until dispatch.

#### What Tenzo Can Learn

✓ Zone-based positioning reduces travel time — Tenzo could position idle experts near high-demand areas during low periods.

# User Peronas

## Who We're Designing For

### Priya S.

**Scheduled Customer · Working professional, 32**

"I booked this a week ago and paid upfront. I need someone here at 9 AM sharp — I have a meeting after."

#### Frustrations

- Last-minute cancellations after she's already planned around it
- Finding out late via app, not a call
- No compensation offered when Tenzo fails

#### Goals

- Guaranteed expert at confirmed time slot
- Proactive communication if anything changes
- Know who's coming before they arrive
- Zero disruption to her day's schedule

#### Success Looks Like

- Expert arrives on time, no surprises, job done.

### Vikram M.

**Instant Customer · Freelancer, 27**

"My place is a mess before my in-laws visit. I need someone now — if it takes more than 30 mins to arrange, I'll figure it out myself."

#### Frustrations

- Blank rejection with no alternative offered
- ETA that keeps shifting after booking
- App shows "confirmed" but expert never shows up

#### Goals

- Fast confirmation — under 3 minutes ideally
- Clear ETA from the moment he books
- Option to see a later slot if now isn't available
- Reliable follow-through once confirmed

#### Success Looks Like

- Instant confirmation or a clear next slot — not silence.

### Aarav K.

**Tenzo Expert · House help professional, 29**

"I work hard all day. I just want to know my schedule in advance so I can plan my travel and not end up rushing between jobs."

#### Frustrations

- Last-minute cancellations after she's already planned around it
- Finding out late via app, not a call
- No compensation offered when Tenzo fails

#### Goals

- Know the day's schedule before leaving home
- Fair distribution of jobs — not always getting the hard ones
- Not penalized if the system assigns a bad job sequence
- Clear in-app guidance when something changes

#### Success Looks Like

- Predictable schedule, fair earnings, no surprise cancellations.

# User Stories

## What Each User Needs the system To Do

Derived from persona frustrations and goals. Each story maps directly to a feature in the assignment engine.

US-01

- As a scheduled customer, I want my expert confirmed at least 30 minutes before my booking time, so that I know someone is definitely coming and can plan my day accordingly.

US-02

- As a scheduled customer, I want to be notified proactively if my expert changes or is running late, so that I'm not waiting at home without information.

US-03

- As a scheduled customer, I want my prepaid booking to be protected from instant orders consuming my expert, so that my advance booking always takes priority over last-minute demand.

US-04

- As a scheduled customer, I want to receive a reschedule option and SLA credit automatically if Tenzo cannot fulfill my booking, so that I feel fairly compensated for the disruption.

US-05

- As an instant customer, I want to know within 60 seconds whether an expert is available now or when the next slot is, so that I can decide quickly without waiting in uncertainty.

US-06

- As an instant customer, I want to be shown a later available ETA instead of a blank rejection, so that I can still book with Tenzo rather than switching to a competitor.

US-07

- As an instant customer, I want real-time tracking of my expert's location once confirmed, so that I don't feel anxious about whether they're actually coming.

US-08

- As an instant customer, I want to be reassigned to another expert automatically if my original expert doesn't start moving within 5 minutes, so that my time isn't wasted on a no-show.

US-09

- As an expert, I want to see my full day's scheduled orders before my shift starts, so that I can plan my travel route and know what to expect.

US-10

- As an expert, I want a clear in-app alert when an instant order is assigned to me, with sufficient travel time built in, so that I'm never expected to teleport between jobs.

US-11

- As an expert, I want my reliability score to only reflect factors within my control, so that system reassignments or scheduling errors don't unfairly lower my rating.

Data Model

The core objects the assignment engine operates on. Not a full ERD — just the fields that matter for the algorithm to run.

Key Entities

| Orders             |                                                          |
|--------------------|----------------------------------------------------------|
| order_id           | Unique identifier                                        |
| type               | INSTANT   SCHEDULED                                      |
| booking_day        | SAME_DAY   PREVIOUS_DAY                                  |
| requested_time     | Timestamp<br>(when service should start)                 |
| duration_mins      | 60   90   120                                            |
| location           | Manikonda   Narsingi   Puppalaguda                       |
| status             | PENDING   ASSIGNED   IN_PROGRESS<br>COMPLETE   CANCELLED |
| assigned_expert_id | FK → Expert                                              |
| priority_score     | Computed at assignment time                              |
| protection_window  | Computed: requested_time - dynamic window                |

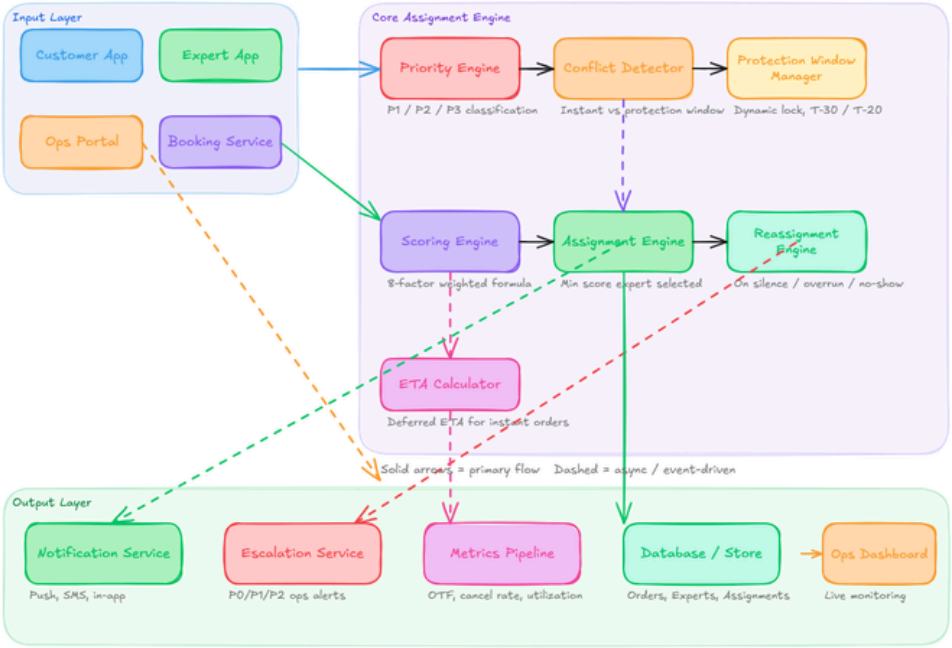
| Assignment      |                                       |
|-----------------|---------------------------------------|
| assignment_id   | Unique identifier                     |
| order_id        | FK → Order                            |
| expert_id       | FK → Expert                           |
| assignment_sco  | Final score at time of assignment     |
| travel_start    | When expert begins traveling to order |
| service_start   | Arrival + service begin               |
| service_end     | Service completion time               |
| buffer_ends     | Expert available again after          |
| is_reassignment | Boolean — was this a fallback?        |

| Expert                  |                                                                |
|-------------------------|----------------------------------------------------------------|
| expert_id               | Unique identifier                                              |
| name                    | Display Name                                                   |
| home_location           | Starting zone for shift                                        |
| current_location        | Live GPS zone (updated every 2 min)                            |
| shift_start / shift_end | 8:00 AM / 6:00 PM                                              |
| status                  | AVAILABLE   IN_TRANSIT   IN_SERVICE<br>  PROTECTED   OFF_SHIFT |
| jobs_today              | Count of completed orders this shift                           |
| idle_mins_today         | Total unassigned time in shift                                 |

| Escalation Event |                                                                 |
|------------------|-----------------------------------------------------------------|
| event_id         | Unique identifier                                               |
| order_id         | FK → Order                                                      |
| trigger_type     | UNASSIGNED_T20   EXPERT_SILENT<br>  OVERRUN   NO_ALTERNATE      |
| severity         | P0   P1   P2                                                    |
| triggered_at     | Timestamp                                                       |
| resolved_at      | Timestamp (null if open)                                        |
| resolution       | REASSIGNED   CUSTOMER_NOTIFIED<br>  MANUAL_OVERRIDE   CANCELLED |

# System Architecture

## How the Components Connect



## Priority Logic

| Order Type             | Priority      | Reasoning                                                                                                                                                                                                                          |
|------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Previous-day scheduled | P1-Highest    | Customer paid in advance with maximum time-certainty expectation. Cancellation causes the most trust damage and likely a refund obligation. Protected at nearly all costs – only displaced if a confirmed alternate expert exists. |
| Same-day scheduled     | P2 - High     | Prepaid and confirmed, but booked closer to service time – slightly more flexibility for reassignment. Still fully protected. No instant order should displace this without a confirmed alternate.                                 |
| Instant order          | P3 - Standard | Demand-driven, real-time. Accepted only when it does not compromise any scheduled order's protection window. Speed matters, but not at the expense of pre-committed bookings.                                                      |



When Can Instant Override a Future Scheduled?

Only when ALL four conditions are simultaneously true:

- **Condition 1:** Time to scheduled start  $\geq 90$  minutes
  - **Condition 2:** Expert finishes Instant (travel + service + buffer) BEFORE scheduled protection window opens
  - **Condition 3:** At least 1 other expert is CONFIRMED available for the scheduled order
  - **Condition 4:** Accepting instant does NOT reduce scheduled OTF probability below 97%
- **Accept Instant only if:**  $\text{current\_time} + 10 + \text{instant\_duration} + 10 + 15 < \text{protection\_window\_start}$

## Protection Window Design

### When to Lock an Expert for a Scheduled Order

| Design Decision                       | Choice             | Rationale                                                                                                                                                                         |
|---------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fixed or dynamic?                     | Dynamic            | Travel time varies — a fixed 30-min window over-protects 0-travel orders and under-protects 12-min travel orders. Base window = pre_travel + 10 min prep buffer, minimum 20 min.  |
| Location-dependent?                   | Yes                | $\text{window\_start} = \text{service\_time} - \text{travel\_time\_from\_nearest\_expert} - 10 \text{ min}$ . Floor: 20 min regardless of travel.                                 |
| Previous-day vs. same-day?            | Prev-day-larger    | Previous-day: T-40 min. Same-day: T-30 min. Higher trust commitment on prev-day warrant more recovery margin.                                                                     |
| Expert availability dependent?        | Yes                | If only one expert can serve an order (proximity or skill), window expands to T-45 min — more margin for failure recovery.                                                        |
| Demand Forecast dependant?            | Yes                | During high-demand periods (weekday 8–10 AM), add 10 min to window — instant traffic is highest and most likely to consume available experts.                                     |
| Should 4 PM order block from 3:30 PM? | Yes, conditionally | 3:30 PM is the minimum start (30 min before). If expert is already at the order location, window can start at T-20. If no expert locked by 3:30, escalation triggers immediately. |

## Three-Stage Escalation When No Expert Found

### 1. T-30: Assign + Lock

Assign best-scored available expert. Freeze them from instant order pool. Customer gets "Expert confirmed" notification.

### 2. T-20: Ops alert if still unassigned

Ops team gets PI alert. Customer gets "We're confirming your expert — update in 5 min" message. System expands search radius by 20%.

### 3. T-10: Manual override + customer call

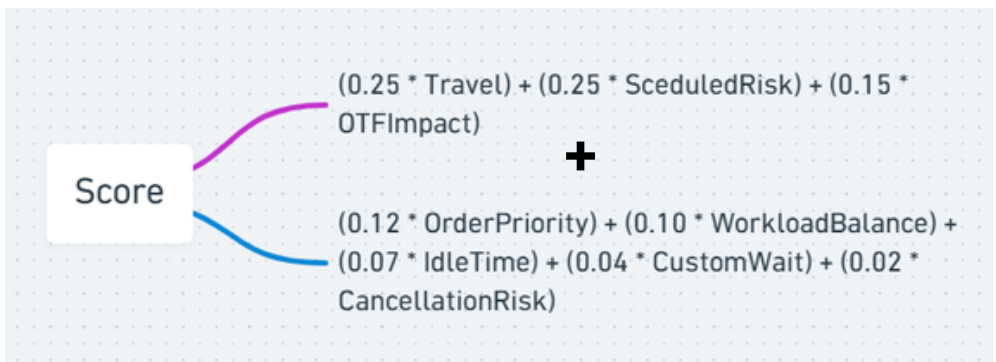
Ops manually assigns or escalates. Customer gets a direct call — offered: accept 10–15 min delay, or reschedule for free + SLA credit. Never a silent cancellation.

#### The 3:29 PM Scenario (O11 + O12)

O12 scheduled 4:00 PM (90 min, Narsingi). Protection window opens at 3:30 PM. O11 arrives at 3:29 PM (60 min, Manikonda). Instant completion = 3:29 + 10 + 60 + 10 + 15 = 4:54 PM. This fully overlaps O12's window. Even if O12 has a separate confirmed expert, O11 still uses up the entire available pool during O12's active service. Decision: REJECT O11. Message: "All experts near you are reserved for upcoming bookings. Try again after 4:00 PM."

## Expert Scoring Formula

How we pick the best expert



All 8 factors normalized 0–100. Score range: 0 (perfect) → 100 (worst). Assign minimum-score expert. Total weights = 1.0.

| Factor                      | Weight | Score Logic                                                                                                                                                   |
|-----------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Distance / Travel time      | 0.25   | Score = (travel_mins / max_travel_in_pool) × 100                                                                                                              |
| Future scheduled order risk | 0.25   | Score = 100 × e <sup>(-gap_mins/30)</sup> . Zero if no scheduled order within 3 hrs.                                                                          |
| OTF impact                  | 0.15   | Score = (1 - expert_30d_OTF_rate) × 100. An expert with 90% OTF history scores 10; one with 70% OTF scores 30.                                                |
| Order priority              | 0.12   | P1 (prev-day scheduled) = 0 penalty to best expert. P2 (same-day scheduled) = 20. P3 (instant) = 50. Score = priority_penalty × (1 - expert_reliability/100). |
| Expert jobs completed today | 0.1    | Score = (jobs_today / max_jobs_in_pool) × 100                                                                                                                 |
| Expert idle time            | 0.07   | Score = 100 - (idle_mins_today / shift_duration × 100)                                                                                                        |
| Customer wait time          | 0.04   | Score = (current_ETA_mins / SLA_threshold_mins) × 100                                                                                                         |
| Cancellation risk           | 0.02   | Score = expert_30d_cancellation_rate × 100                                                                                                                    |

For Scheduled orders: ScheduledRisk weight = 0.0 (this IS the order being assigned). OTFImpact rises to 0.25, OrderPriority to 0.20, Travel to 0.30, Workload to 0.15, Idle to 0.10. For Instant orders: use the full 8-factor formula above — Travel and ScheduledRisk dominate since speed and conflict avoidance matter most. This ensures instant assignment optimizes for speed while scheduled assignment optimizes for reliability, fairness, and OTF history.

# Sceduling Exercise

## All 10 Orders Assigned

| Order  | Type     | Expert      | Travel   | Arrival  | Service start | Service End | Post Travel Ends | Buffers Ends |
|--------|----------|-------------|----------|----------|---------------|-------------|------------------|--------------|
| Rahul  | P1 sched | E1 – Aarav  | 8:00 AM  | 8:10 AM  | 8:10 AM       | 9:10 AM     | 9:20 AM          | 9:35 AM      |
| Priya  | P1 Sched | E2 – Bhavya | 8:00 AM  | 8:10 AM  | 8:10 AM       | 10:10 AM    | 10:20 AM         | 10:35 AM     |
| Ankit  | Instant  | E4 – Divya  | 8:05 AM  | 8:15 AM  | 8:15 AM       | 9:15 AM     | 9:25 AM          | 9:40 AM      |
| Sneha  | P1 Sched | E3 – Charan | 9:10 AM  | 9:22 AM  | 9:30 AM       | 10:52 AM    | 11:02 AM         | 11:17 AM     |
| Vikram | Instant  | E4 – Divya  | 9:40 AM  | 9:50 AM  | 9:50 AM       | 11:20 AM    | 11:30 AM         | 11:45 AM     |
| Neha   | P1 Sched | E1 – Aarav  | 10:45 AM | 10:45 AM | 11:00 AM      | 12:00 AM    | 12:10 AM         | 12:25 AM     |
| Rohan  | Instant  | E4 – Divya  | 11:45 AM | 11:57 AM | 11:57 AM      | 1:57 PM     | 2:07 PM          | 2:22 PM      |
| Meera  | P2 Sched | E2 – Bhavya | 12:00 PM | 12:10 PM | 12:30 PM      | 2:00 PM     | 2:10 PM          | 2:25 PM      |
| Aisha  | P1 Sched | E1 – Aarav  | 1:30 PM  | 1:40 PM  | 2:00 PM       | 4:00 PM     | 4:10 PM          | 4:25 PM      |
| Karan  | Instant  | E3 – Charan | 2:20 PM  | 2:32 PM  | 2:32 PM       | 3:32 PM     | 3:42 PM          | 3:57 PM      |

INSTANT ORDER DECISION LOGIC

Accept, Reject, or Show Later ETA

| Order                                          | Type      | Expert      | Travel   | Arrival                                                                                                                                                                                           |
|------------------------------------------------|-----------|-------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| O3 Ankit<br>8:05 AM ·<br>Manikonda · 60 min    | Accept    | E4 – Divya  | 8:15 AM  | E1 on O1. E4 free at Manikonda, 0 travel. Completes 9:25 AM — no scheduled conflict. Accept immediately.                                                                                          |
| O5 Vikram<br>8:20 AM ·<br>Narsingi · 90 min    | Later ETA | E4 – Divya  | 9:50 AM  | No expert free at Narsingi. E4 free at 9:40, +10 min travel. Show: "Next available: 9:50 AM — confirm or cancel." No conflict with O8 (E2).                                                       |
| O7 Rohan<br>9:00 AM ·<br>Puppalaguda · 120 min | Later ETA | E4 – Divya  | 11:52 AM | E3 locked for O4. E4 free at 11:45 after O5, +12 min travel. Show: "Next available: ~11:52 AM." No conflict in afternoon window.                                                                  |
| O10 Karan<br>2:30 PM ·<br>Puppalaguda · 60 min | Accept    | E3 – Charan | 2:32 PM  | E3 free since 11:17 AM. No afternoon scheduled conflicts. 0 travel. Accept immediately.                                                                                                           |
| O11 Sanjana<br>3:29 PM ·<br>Manikonda · 60 min | Reject    | N/A         | N/A      | O12 (4:00 PM) protection window opens at 3:30 PM — 1 min away. Instant completion = 4:54 PM. Fully overlaps O12 window. Reject with: "Experts reserved for upcoming bookings. Try after 4:00 PM." |

# EDGE CASES & DECISION FLOWS

## Five Scenarios the System Must Handle

### Scenario 1 — New Scheduled Booking at 10:17 AM While Expert Is En Route for Instant

1.Immediately soft-lock the 11:30 AM slot on payment

Do not wait for the standard protection window. Mark an expert as pending-confirmation the moment payment clears.

2.Check if in-progress expert can make 11:10 AM

10:17 + remaining service + 10 post + 15 buffer ≤ 11:10? If yes → keep instant running, assign same expert to scheduled.

3.If not → find alternate for scheduled, keep instant running

Never cancel an in-progress service. Scan all other experts. Assign the best-scored alternate to the 11:30 booking.

4.If no alternate → notify new customer proactively

"Confirming your expert — update in 5 min." Ops P1 alert. If unresolved by T-15: direct call, offer reschedule + credit.

### Scenario 2 — 60-Min Order Running 75 Minutes, Expert Has Scheduled Order After

1.Flag at +10 min overrun mark

System detects service running over expected duration. Check: does extended end time breach next scheduled's buffer?

2.If buffer intact → no action

Absorb within existing buffer. Next order is safe. No customer notification needed.

3.If buffer breached → reassign next scheduled order

Auto-initiate reassignment. Notify next customer: "Your expert has been updated to [Name] — no change to your service time." Frame as upgrade.

4.If no alternate → offer customer choice

Accept 10–15 min delay, or reschedule free + credit. Never silently show up late.

### Scenario 3 — Expert A (200m closer, 6 jobs) vs Expert B (farther, 1 job)

| Condition                   | Assign To | Reason                                                                                                                                                                           |
|-----------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Instant Order, SLA urgency  | Expert A  | Travel time varies — a fixed 30-min window over-protects 0-travel orders and under-protects 12-min travel orders. Base window = pre_travel + 10 min prep buffer, minimum 20 min. |
| Scheduled order, not urgent | Expert B  | Workload balance has higher weight for scheduled. Expert A at 6 jobs risks fatigue-driven quality degradation. Extra travel is worth the balance benefit.                        |

Scenario 4 — One Expert, One Instant Now, One Scheduled in 35 Minutes

1. Calculate instant completion time

For a 60-min instant: 10 + 60 + 10 + 15 = 95 min total. Expert free at T+95, well past the 35-min window.

2. Instant cannot fit → decline or defer

Show instant customer: "Next available slot: after [scheduled end time]." Prepaid scheduled customer always takes priority with one expert.

Scenario 5 — Expert Accepts But Doesn't Move Within 3 Minutes

1. T+3: Push notification to expert

"Confirm you're on the way" — one-tap: "On my way" or "I have an issue."

2. T+5: No response → auto-reassign

Customer notified: "Your expert has been updated." Original expert account flagged for review.

ALL FLOWS IN ONE PLACE

| Flow                    | trigger                                                             | Logic                                                                                                                                                                                 |
|-------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reassignment            | Expert offline, silent 5+ min, or 10+ min overrun                   | Re-run scoring on remaining experts. Must arrive within acceptable ETA. Customer notified within 30 seconds. If no alternate: ops alert → customer call → reschedule offer.           |
| Extension               | Expert taps "need more time" or system auto-detects overrun via GPS | Check if extension breaches next scheduled buffer. If no → silently approve, adjust ETA. If yes → immediate reassignment for next order.                                              |
| Cancellation Prevention | Scheduled order unassigned approaching T-30                         | T-30: assign + lock. T-20: ops P1 alert + customer message. T-10: ops manual override + customer call with choice of delay or reschedule + credit.                                    |
| Customer Communication  | Every status change                                                 | Instant accepted → push + in-app. Deferred ETA → SMS + in-app confirmation request. Reassignment → push + SMS within 30s. Possible delay → push + call at T-10.                       |
| Admin Escalation        | System trigger by severity                                          | P1: shift ops lead (resolve in 10 min). P0: ops manager + product on-call (immediate war room). Auto-trigger: 3+ cancellations/hour, OTF <90% in last 2 hrs, expert silent mid-order. |

TRADE-OFF DECISIONS

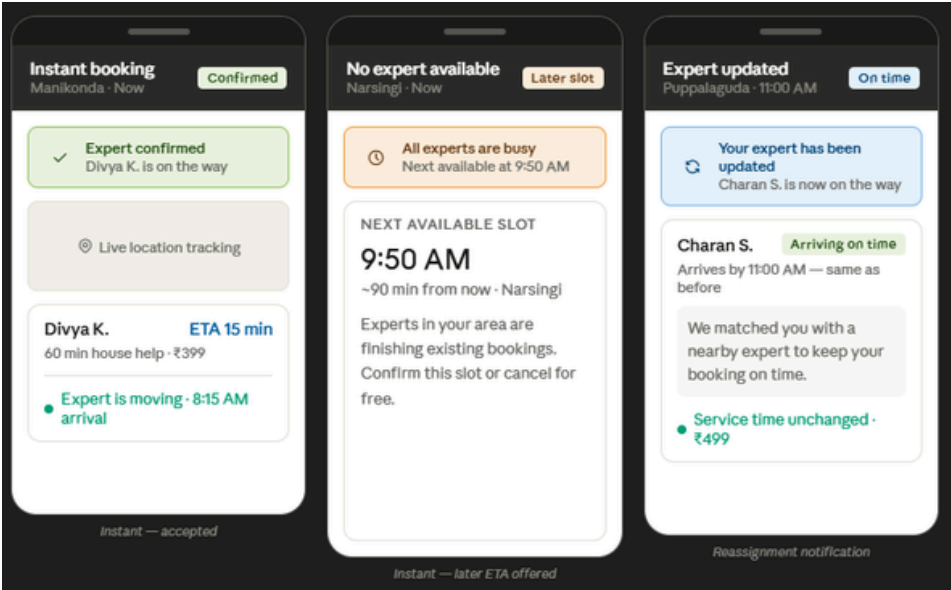
| Decision                                                  | What we chose                                                   | What we Rejected                               | Trade-Off Accepted                                                                                                                                                                                                    |
|-----------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Protection window: fixed vs dynamic                       | Dynamic (location + demand aware)                               | Fixed 30-min window for all orders             | More compute complexity. Requires live location data. But: fixed window over-protects 0-travel orders and under-protects 12-min travel ones — hurts instant acceptance rate unnecessarily.                            |
| Instant rejection: hard reject vs deferred ETA            | Show later ETA, ask customer to confirm                         | Hard reject with no alternative                | Customer may not return for deferred slot. But: hard reject loses the booking entirely. Deferred ETA converts potential lost demand into future revenue — even 30% conversion is better than 0%.                      |
| In-progress instant order conflict: cancel vs alternate   | Never cancel in-progress. Always find alternate for next order. | Cancel instant if scheduled order at risk      | Finding an alternate is harder and sometimes impossible. But: canceling a customer mid-service destroys trust twice. The complexity cost is worth the trust preservation.                                             |
| Expert pool: shared vs separate for instant and scheduled | Shared pool with conflict resolution engine                     | Separate dedicated pools (Urban Company model) | Shared pool is harder to manage but significantly more capital-efficient. Separate pools solve the conflict but create idle inventory on both sides. Tenzo's unit economics demand a shared pool — hence this engine. |



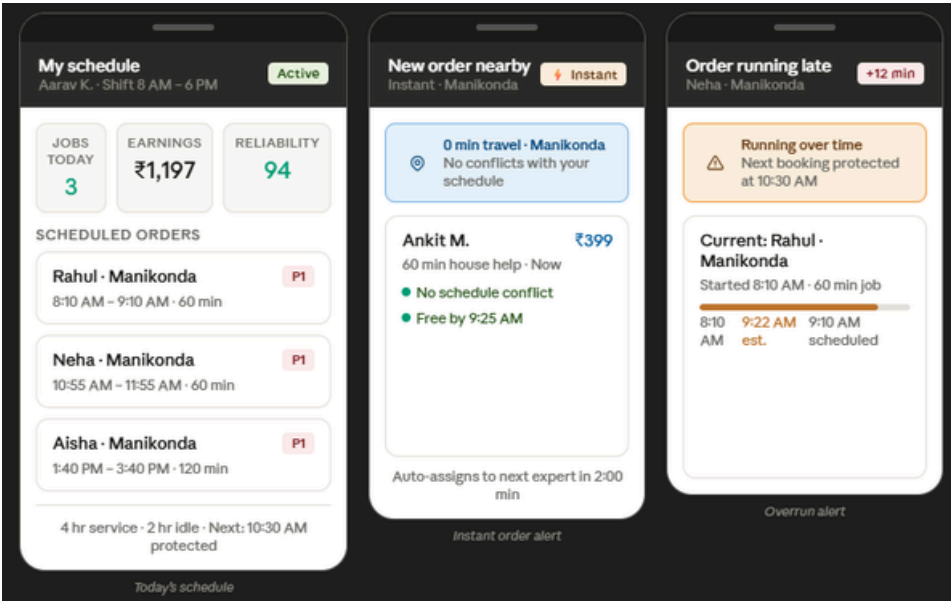
RISK REGISTER

| Risk                                                                                                                         | Likelihood | Impact                                         | Mitigation                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expert GPS spoofing<br>Expert marks as "moving" without actually leaving                                                     | Medium     | Fixed 30-min window for all orders             | More compute complexity.<br>Requires live location data. But: fixed window over-protects 0-travel orders and under-protects 12-min travel ones — hurts instant acceptance rate unnecessarily.                         |
| Zone demand spike exhausts all experts<br>Flash demand in one area leaves others unserved                                    | Medium     | Hard reject with no alternative                | Customer may not return for deferred slot. But: hard reject loses the booking entirely.<br>Deferred ETA converts potential lost demand into future revenue — even 30% conversion is better than 0%.                   |
| Two instant orders arrive simultaneously for same expert<br>Race condition in assignment logic                               | Low        | Cancel instant if scheduled order at risk      | Finding an alternate is harder and sometimes impossible. But: canceling a customer mid-service destroys trust twice. The complexity cost is worth the trust preservation.                                             |
| Protection window too aggressive → kills instant acceptance<br>Dynamic window keeps experts locked too long                  | Medium     | Separate dedicated pools (Urban Company model) | Shared pool is harder to manage but significantly more capital-efficient. Separate pools solve the conflict but create idle inventory on both sides. Tenzo's unit economics demand a shared pool — hence this engine. |
| Expert reliability score drops due to system error<br>System-initiated reassignment unfairly penalizes expert                | Medium     | Medium                                         | Tag every reassignment with trigger type (EXPERT_FAULT vs SYSTEM_FAULT). Only EXPERT_FAULT events affect reliability score. System-initiated reassignments excluded from scoring.                                     |
| Customer books scheduled, immediately instant arrives for same expert<br>Edge case where protection window hasn't propagated | Low        | Medium                                         | Protection window written synchronously at booking time, before any instant order can read expert availability. Scheduled booking creation and window lock are a single atomic transaction.                           |

CUSTOMER APP



EXPERT APP



# SOLUTION DESIGN

## How The System Works End-to End

### When a Scheduled Order Arrives

#### 1. Classification

System reads booking\_day (prev-day = P1, same-day = P2) and writes a ProtectionWindow record. Window start = service\_time - dynamic\_travel - 10 min. Floor: 20 min.

#### 2. Pre-Assignment

At protection\_window\_start, scoring engine evaluates all non-protected experts. Minimum-score expert gets locked. Expert status → PROTECTED. Instant order pool no longer sees this expert.

#### 3. Customer Confirmation

Expert confirmed push sent at T-30. If no expert locked by T-20, escalation triggers. Customer notified proactively — never discovers a cancellation on their own.

### When an Instant Order Arrives

#### 1. Conflict Check (mandatory first step)

System reads all active ProtectionWindows. For every available expert, calculates: current\_time + 10 + instant\_duration + 10 + 15. Any expert where this value overlaps a protection window is removed from consideration.

#### 2. If experts remain → score and assign

Scoring formula runs on remaining experts. Minimum score wins. Assignment written. Expert status → IN\_TRANSIT. Customer gets ETA.

#### 3. If no experts remain → calculate next available ETA

System finds the earliest time any expert completes their current assignment + buffer. Shows customer that ETA with confirm / cancel option. Never a blank rejection

#### 4. If deferred ETA is confirmed → queue the order

Order enters PENDING state with a future assignment time. At that time, conflict check runs again (conditions may have changed) and the best expert is assigned.

# ACCEPATANCE CRITERIA

## AC-01 - Scheduled Order Protection

**Given** A scheduled order exists for 4:00 PM and no expert is assigned  
**When** The clock reaches T-30 (3:30 PM)  
**Then** The system must assign a qualified expert and set their status to PROTECTED. No instant order can claim this expert until the service and buffer are complete.

**AC-02 · Instant Order Conflict Check**

**Given** A scheduled order has a protection window from 3:30 PM to 5:00 PM  
**When** An instant order arrives at 3:29 PM requiring 60 min service  
**Then** The system must calculate completion time (3:29 + 95 min = 4:54 PM), identify the overlap, and reject the instant order with a message showing the next available slot after 5:00 PM.

**AC-03 · Deferred ETA Offer**

**Given** No expert is available for an instant order right now  
**When** The system determines the earliest available expert slot  
**Then** The customer must see a "Next available at [time]" offer with two options: confirm the deferred slot, or cancel at no charge. The system must not show a generic rejection.

**AC-04 · Expert Non-Start Handling**

**Given** An expert has accepted an order  
**When** 5 minutes pass with no GPS movement detected  
**Then** The system auto-reassigns to the next best available expert, notifies the customer within 30 seconds, and flags the original expert's account. The customer must not experience more than 7 minutes of unresolved uncertainty.

**AC-05 · Escalation at T-20**

**Given** A scheduled order is 20 minutes from its start time  
**When** No expert has been assigned or confirmed  
**Then** A PI alert must fire to the shift ops lead. The customer must receive a proactive message ("confirming your expert — update in 5 min") before they reach out to support.

**AC-06 · Workload Balance**

**Given** Expert A has completed 7 jobs today; Expert B has completed 2 jobs, is 3 min farther  
**When** A new scheduled order arrives  
**Then** Expert B must be assigned. The scoring formula must produce a lower score for B due to workload balance weight, overriding A's proximity advantage for scheduled orders.

**AC-07 · Overrun Extension**

**Given** An expert's current service is running 10+ minutes over scheduled duration  
**When** The overrun would breach the buffer before the expert's next scheduled order  
**Then** The system must immediately find an alternate expert for the next order, not wait for the overrun to continue. The next customer must receive an "expert updated" notification before their original ETA passes.

# PRODUCT METRICS & IMPROVEMENT PLAN

## Cancellation Rate

**18%**

**Target: <5%**

### Product fix

Protection windows + pre-assignment at T-30. Proactive escalation at T-20. Deferred ETA converts rejections to queued bookings instead of lost orders.

### Algorithm change

Add ScheduledRisk factor to scoring formula (weight 0.25). Any expert whose completion time overlaps a protection window is removed from candidate pool before scoring runs — hard filter, not just a penalty. Protection window floor increased from 20 min to 25 min for prev-day orders during peak hours (8–10 AM).

## On-Time Fulfillment

**84%**

**Target: >95%**

### Product fix

Dynamic buffers + location-aware protection windows. GPS monitoring alerts at T-10 if expert not moving. Overrun detection at +10 min triggers

### Algorithm change

Add OTF Impact factor (weight 0.15) to scoring formula — experts with lower 30-day OTF history score higher (worse) and are deprioritized for P1 orders. Buffer time for scheduled orders increased from 15 min to 18 min when expert's last location is >10 min from next order, to absorb real-world travel variance.

## Scheduled Fulfillment

**88%**

**Target: >97%**

### Product fix

Prev-day orders locked at T-40, same-day at T-30. Ops alert at T-20. Never let a scheduled order reach T-10 unassigned.

### Algorithm change

Order Priority factor (weight 0.12) added to scoring formula — P1 orders pull the highest-reliability experts by design. Protection window assignment now runs on a recurring 5-minute sweep (not just triggered at a single threshold), so if an expert becomes available earlier than expected, the system can assign ahead of schedule and close the window sooner. Reduces T-20 breach rate.

## On-Time Fulfillment

**52%**

**Target: >80%**

### Product fix

Deferred ETA offer replaces hard rejection. Customer sees next available slot with confirm/cancel choice. Track deferred-to-confirmed conversion rate as a sub-metric.

### Algorithm change

Protection window conflict check now runs as a pre-filter before the full scoring formula — only truly conflicting experts are excluded, not all busy experts. Previously, system excluded any expert not marked AVAILABLE. New logic: expert is eligible if  $\text{current\_order\_completion} + \text{travel} \leq \text{instant\_service\_start}$ . This alone is expected to recover 15–20%.

METRICS DASHBOARD DESIGN

| Panel                | Metrics Shown                                                                                                       | Alert Threshold                   | Action Enabled                                    |
|----------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------|---------------------------------------------------|
| Live Order Map       | Expert locations, order status (in-transit, in-service, idle, protected), ETA accuracy vs. promised                 | Expert idle >30 min               | Manual reassignment, zone rebalancing             |
| Fulfillment Health   | OTF % (rolling 1 hr + 24 hr), Cancellation rate, Scheduled fulfillment %, Protection window breach rate             | OTF <90% → P1 alert               | Expand search radius, escalate to manager         |
| Instant Order Funnel | Received → Accepted → Deferred → Confirmed Deferred → Rejected. Rejection reason breakdown (protection conflict vs. | Rejection rate >30%               | Review protection window parameters               |
| Expert Status Board  | Each expert: current order, jobs today, idle time today, reliability score, next scheduled order                    | Expert >8 jobs → soft cap alert   | Cap expert from new instant orders                |
| Escalation Queue     | All open escalations by severity (P0/P1/P2), time since trigger, order details, available alternates                | Any P1 open >5 min → manager ping | Assign manually, call customer, offer credit ping |
| Root Cause Tracker   | Cancellation reason breakdown: expert-fault / system-fault / customer-fault. Trending patterns over 7 days.         | System-fault >5% of cancellations | Ops review of algorithm parameters                |

# GO-TO-MARKET PLAN

## Phase 1 · Weeks 1–3

### Shadow Mode

New assignment engine runs in parallel with the existing system. Old engine makes all real decisions. New engine's decisions are logged but not executed. No customer or expert impact.

Goal: measure disagreement rate between old and new engine. Identify edge cases the algorithm handles poorly before they affect real orders.

Exit criterion: New engine agrees with human ops decisions >90% of the time on a labelled set of 200 historical orders.

## Phase 2 · Weeks 4–6

### Scheduled Orders Only — One Zone

New engine goes live for scheduled order assignment in Manikonda only. Instant orders still use old logic. Ops team monitors protection window breach rate and escalation queue daily.

Goal: validate that protection window logic reduces T-20 unassigned events. Gather ops feedback on escalation flow usability.

Exit criterion: Scheduled fulfillment rate in Manikonda  $\geq 95\%$  for 2 consecutive weeks. Zero unexpected cancellations caused by new engine.

## Phase 3 · Weeks 7–10

### Full Rollout — All Zones, Instant Logic Enabled

New engine live for all order types across all zones. Instant conflict detection and deferred ETA offers go live. A/B test: 50% of instant rejections show deferred ETA offer, 50% get hard reject (control). Measure conversion of deferred ETAs.

Expert reliability scoring goes live. Experts can see their scores but no consequences applied for first 2 weeks — grace period.

Exit criterion: Instant acceptance rate (accepted + deferred confirmed)  $\geq 70\%$ . Cancellation rate trending below 10%.

## Phase 4 · Weeks 11–13

### Optimization + Dashboard Live

Ops dashboard deployed. Expert reliability score consequences activated. A/B test results read — winning instant order approach standardized. Dynamic protection window parameters tuned based on 4 weeks of live data.

Retrospective with ops team. Engineering review of P0/P1 incidents from Phases 2–3. Roadmap for Phase 2 features (skill-based routing, live traffic) drafted based on actual pain points observed.

Exit criterion: All 5 metrics at or trending toward 90-day targets. Dashboard adopted by ops team as primary monitoring tool.

### Rollback Plan

If cancellation rate spikes above 20% at any point in Phase 3 or 4, revert to old assignment engine within 1 hour — feature flag flip, no deployment needed. All Assignment records from new engine retained for post-mortem analysis. Customer-facing impact should be zero — old engine picks up from current expert states.